

Design and Implementation of a Wazuh-Based SOC Home Lab for Attack Detection and Log Analysis

Comprehensive Technical Investigation into Multi-Platform Telemetry, eBPF Kernel Monitoring, Cyber Attack Detection Engineering, and SIEM Log Analysis

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Platform Stack:	Wazuh 4.x (Manager/Indexer/Dashboard), Microsoft Sysmon (Windows & Linux eBPF)
Virtualization:	Oracle VirtualBox Isolated NAT Network (10.0.2.0/24 Subnet)
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EXECUTIVE ABSTRACT:

*This capstone report documents the architecture, deployment, offensive validation, and defensive detection engineering of a robust Security Operations Center (SOC) home laboratory. Utilizing the open-source Wazuh SIEM ecosystem integrated with Microsoft Sysmon for Windows and eBPF-driven Sysmon for Linux, this environment achieves comprehensive visibility across host process lifecycles, network sockets, user authentication, and system configurations. Multiple real-world attack vectors (Nmap network reconnaissance, Hydra brute-force authentication, PowerShell execution, and unauthorized privileged commands) were executed from Kali Linux and correlated in real time. Over **12,600+ telemetry events** were successfully collected, categorized by custom decoders/rules, and mapped against the MITRE ATT&CK; enterprise matrix.*

1. Introduction, Objectives & Problem Statement

In modern enterprise IT environments, security visibility is often severely compromised by fragmented logging mechanisms, high alert fatigue, and lack of host-level process context. Standard operating system event logs (such as generic Windows Security Events or standard Linux syslogs) typically record high-level authentication summaries without providing granular process lineage (Parent Process ID, command-line arguments, process GUIDs, or binary hashes).

To address these challenges, this project implements a practical, cost-effective Security Operations Center (SOC) environment that unifies:

- **Centralized SIEM Management:** Wazuh Manager for log ingestion, decoder parsing, and alert correlation.
- **High-Performance Indexing:** Wazuh Indexer (OpenSearch) for multi-field search and aggregate analytical dashboards.
- **Deep Windows Telemetry:** Microsoft Sysmon capturing process creation, network bindings, and registry modifications.
- **Kernel-Level Linux Telemetry:** Sysmon for Linux (``sysmonforlinux``) leveraging eBPF tracepoints without kernel instability.
- **Empirical Threat Validation:** Controlled attack simulations from Kali Linux to test and tune detection rules.

2. Lab Architecture & Network Topology

The entire laboratory is virtualized on Oracle VirtualBox using an isolated NAT Network (``10.0.2.0/24``), enabling high-speed communication while isolating traffic from the production home network.

Host / Node	Operating System	IP Address	Hardware Specs	Primary Role in Lab
Wazuh Server	Ubuntu Server 24.04 LTS	10.0.2.15	4 GB RAM, 2 vCPU, 50 GB	SIEM Manager, Indexer & Dashboard
Victim Endpoint	Windows 11 Pro (64-bit)	10.0.2.20	4 GB RAM, 2 vCPU, 60 GB	Monitored Endpoint (Agent + Sysmon)
Attacker Node	Kali Linux (Rolling)	10.0.2.30	2 GB RAM, 2 vCPU, 30 GB	Offensive Recon & Attack Platform
Host Machine	Windows 11 Physical	Gateway / Host	16 GB RAM, Core i7	VirtualBox Hypervisor Host

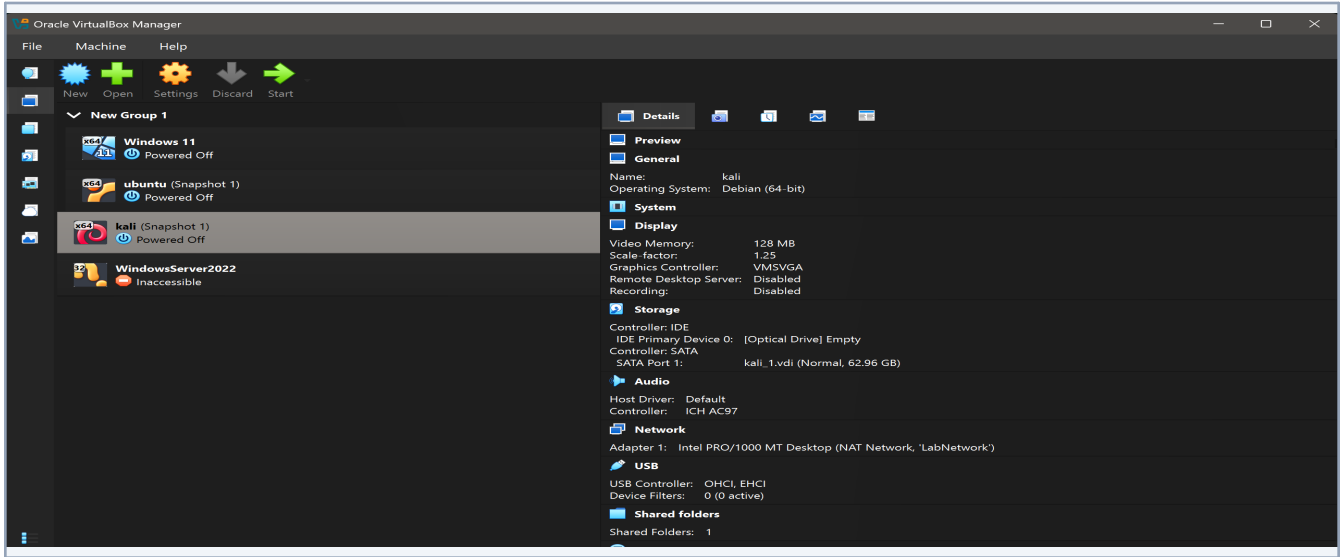


Figure: Oracle VirtualBox Environment Hosting Ubuntu Server, Windows 11, and Kali Linux

3. Wazuh SIEM Deployment & Dashboard Architecture

The Wazuh SIEM platform operates through three tightly coupled subsystems on the Ubuntu 24.04 host:

1. **Wazuh Manager** (`wazuh-manager`): Listens on TCP port 1514 for TLS-encrypted agent communication and port 1515 for dynamic agent registration. Decodes incoming logs and matches them against XML correlation rules in real time.
2. **Wazuh Indexer** (`wazuh-indexer`): Powered by OpenSearch, it indexes incoming security events into structured document shards (`wazuh-alerts-*` and `wazuh-archives-*`), supporting instant full-text and field-specific filtering.
3. **Wazuh Dashboard** (`wazuh-dashboard`): A modern web interface running over HTTPS (TCP port 443) providing Security Events overview, MITRE ATT&CK; visualization, Threat Hunting panels, and Regulatory Compliance reports.

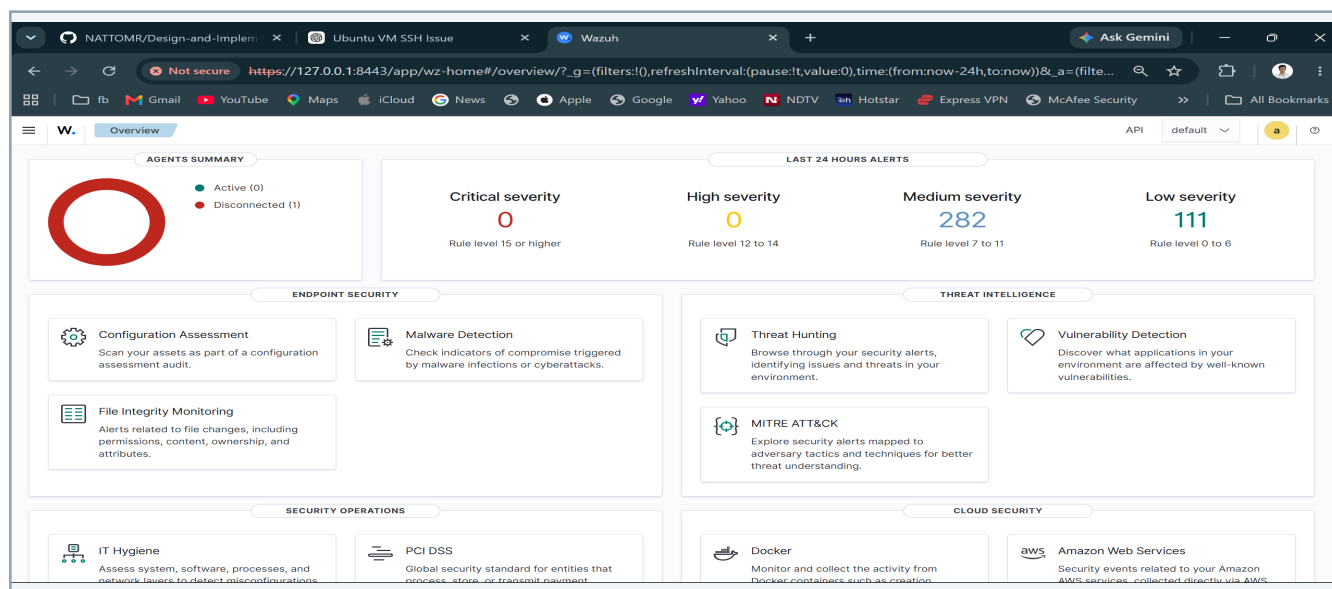


Figure: Wazuh Dashboard Overview Presenting Real-Time Security Events and System Status

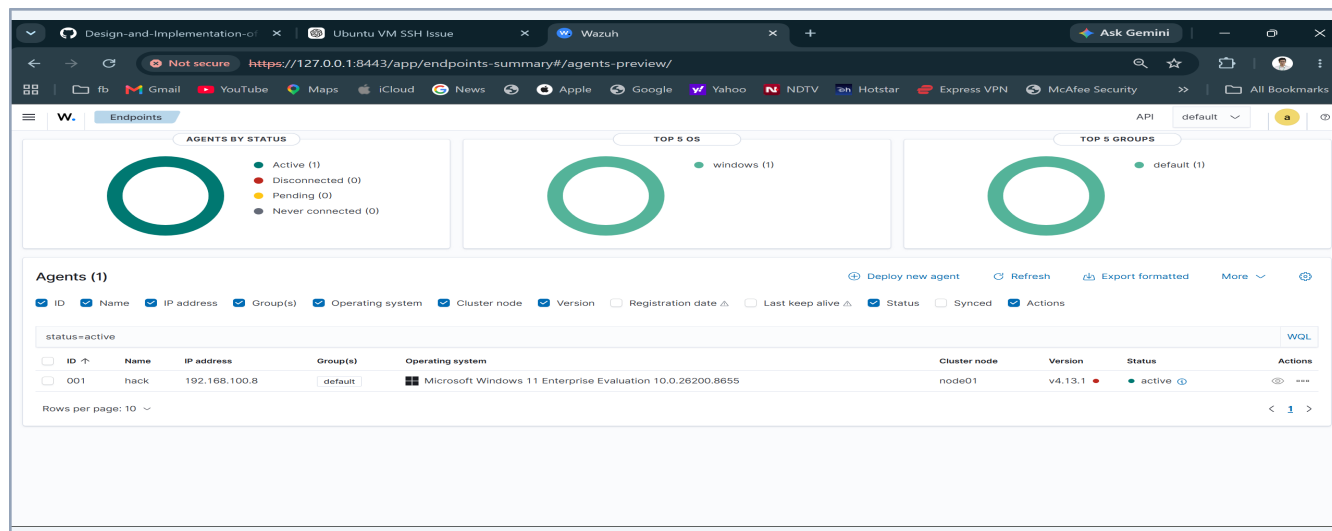


Figure: Wazuh Manager Active Multi-Platform Agents Summary (Windows 11 and Kali Linux)

4. Endpoint Telemetry & Sysmon Integration (Windows & Linux)

Microsoft Sysmon provides high-fidelity endpoint activity monitoring by hooking deep system routines. In this project, Sysmon was deployed on both Windows 11 and Linux to establish unified, cross-platform telemetry.

Event ID	Event Name	Target OS	Operational SOC Value & Threat Coverage
1	Process Creation	Win & Linux	Captures full CLI arguments, ParentProcessId, ProcessGuid, hashes, user context.
3	Network Connect	Win & Linux	Records source/destination IP, port, initiating binary, and protocol (TCP/UDP).
5	Process Terminate	Win & Linux	Tracks process lifecycle duration and malware termination behaviors.
9	Raw Access Read	Linux	Detects direct device access (`/dev/sda`, memory scraping, credential theft).
11	File Create	Win & Linux	Detects dropper payloads in `tmp/`, `C:\Windows\Temp`, and startup directories.
12/13	Registry Event	Windows	Monitors Run/RunOnce persistence keys and defensive software disablement.
23	File Delete	Win & Linux	Tracks anti-forensic evidence destruction and ransomware file wiping.

```
PS C:\Sysmon> .\Sysmon64.exe -accepteula -i sysmonconfig-export.xml

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
```

Figure: Sysmon Installation on Windows 11 via PowerShell with Schema Validation

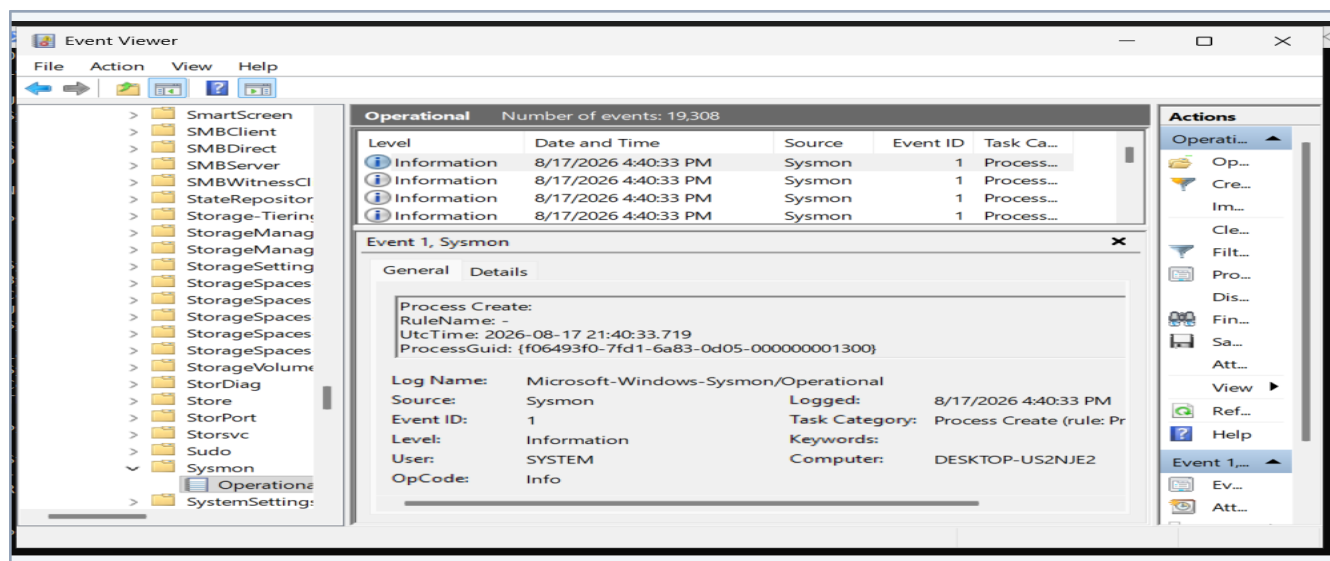


Figure: Windows Event Viewer Displaying Ingested Sysmon Operational Logs (Event ID 1)

5. Critical Command Reference Across All Lab Nodes

To ensure operational efficiency, reproducible administration, and rapid troubleshooting, the most critical commands executed across each lab machine are documented below:

A. Wazuh Server / Ubuntu Commands (Management, Ingestion & Analysis)

Operation Category	Command Line Syntax	Operational Function & Purpose
Service Control	<code>`sudo systemctl status wazuh-manager wazuh-indexer wazuh-dashboard`</code>	Verify status of all core SIEM daemons.
Agent Management	<code>`sudo /var/ossec/bin/agent_control -l`</code>	List all registered, active, and disconnected agents.
Agent Info	<code>`sudo /var/ossec/bin/agent_control -i 003`</code>	Query detailed operating telemetry for agent ID 003.
Rule Testing	<code>`sudo /var/ossec/bin/wazuh-logtest`</code>	Interactive log decoder and rule matching simulator.
Live Alerts Stream	<code>`sudo tail -f /var/ossec/logs/alerts/alerts.json`</code>	Stream raw JSON alert output as events trigger.
Firewall Rules	<code>`sudo ufw allow 1514/tcp && sudo ufw allow 1515/tcp && sudo ufw allow 443/tcp`</code>	Open agent log ingestion, registration & web UI ports.
Manager Logs	<code>`sudo tail -n 100 /var/ossec/logs/ossec.log`</code>	Inspect internal manager errors and parsing issues.

B. Windows 11 Endpoint Commands (PowerShell / CMD as Administrator)

Operation Category	Command Line Syntax	Operational Function & Purpose
Sysmon Install	<code>`cd C:\Sysmon; .\Sysmon64.exe -accepteula -i sysmonconfig.xml`</code>	Install and activate Sysmon system service with XML config.
Sysmon Config Update	<code>`.\Sysmon64.exe -c sysmonconfig.xml`</code>	Reload updated Sysmon XML filtering configuration.
Sysmon Service Check	<code>`Get-Service Sysmon64`</code>	Verify Sysmon service status is in 'Running' state.
Wazuh Service Control	<code>`NET STOP WazuhSvc; NET START WazuhSvc`</code>	Restart Windows Wazuh Agent service to reload configuration.
Query Sysmon Logs	<code>`Get-WinEvent -LogName 'Microsoft-Windows-Sysmon/Operational' -MaxEvents 5`</code>	Read recent local Sysmon events directly via PowerShell.
Query Failed Logons	<code>`Get-WinEvent -FilterHashtable @{LogName='Security';ID=4625}`</code>	Inspect Windows Event 4625 failed login entries.
Connectivity Test	<code>`Test-NetConnection -ComputerName 10.0.2.15 -Port 1514`</code>	Test TCP port reachability to the Wazuh Manager.

C. Kali Linux Attacker & Linux Telemetry Commands

Attack Phase / Role	Command Line Syntax	Offensive & Defensive Objective
Host Discovery	<code>`nmap -sn 10.0.2.0/24`</code>	Perform ICMP/ARP ping sweep to discover live subnet hosts.
TCP Port & OS Scan	<code>`nmap -sS -sV -O -T4 10.0.2.20`</code>	SYN stealth scan to enumerate open ports, services, and OS.
Aggressive Scan	<code>`nmap -A -p 80,443,3389 10.0.2.20`</code>	Execute script scans, traceroute, and version detection.
RDP Brute Force	<code>`hydra -l administrator -P rockyou.txt rdp://10.0.2.20`</code>	Simulate high-frequency credential stuffing against Windows.
Sysmon Linux Service	<code>`sudo sysmon -accepteula -i /etc/sysmon/sysmonconfig.xml`</code>	Activate eBPF kernel telemetry collector on Linux.
Sysmon Linux Logs	<code>`sudo journalctl -u sysmon -f`</code> or <code>`sudo tail -f /var/log/syslog`</code>	Stream live Linux Sysmon process and socket events.

6. Detection Engineering & Custom Rule Authoring

Wazuh features a modular rules engine. To detect threats that evade standard signatures, custom rules were developed in ``/var/ossec/etc/rules/local_rules.xml``:

Implemented Custom Detection Rules (``local_rules.xml``)

```
<group name="custom_rules,sysmon,threat_hunting">
  <!-- Rule 100001: Reconnaissance Port Scanning Detection -->
  <rule id="100001" level="8" frequency="8" timeframe="10">
    <if_matched_group>sysmon_network_connect</if_matched_group>
    <same_source_ip />
    <description>Potential Network Reconnaissance / Rapid Port Scan Detected</description>
    <mitre><id>T1595.001</id></mitre>
  </rule>

  <!-- Rule 100002: Encoded PowerShell & Execution Policy Bypass -->
  <rule id="100002" level="10">
    <if_group>sysmon_process_create</if_group>
    <field name="win.eventdata.commandLine" type="pcre2">(?:i)-enc|-encodedcommand|-exec.*bypass</field>
    <description>Suspicious Encoded or Obfuscated PowerShell CLI Execution</description>
    <mitre><id>T1059.001</id></mitre>
  </rule>

  <!-- Rule 100003: Multiple Logon Failures (Brute-Force Threshold) -->
  <rule id="100003" level="12" frequency="5" timeframe="60">
    <if_matched_sid>18152</if_matched_sid>
    <same_source_ip />
    <description>Potential RDP / Windows Authentication Brute-Force in Progress</description>
    <mitre><id>T1110.001</id></mitre>
  </rule>
</group>
```

MITRE ATT&CK; Matrix Mapping

Tactic	Technique ID	Technique Name	Telemetry Source	Alert Rule Triggered
Reconnaissance	T1595.001	Port Scanning	Sysmon Event ID 3	Rule 100001 (Rapid Port Scan)
Credential Access	T1110.001	Password Guessing	Event ID 4625	Rule 18152 & Rule 100003
Execution	T1059.001	PowerShell CLI	Sysmon Event ID 1	Rule 100002 (Encoded CLI)
Persistence	T1547.001	Registry Run Keys	Sysmon Event 12/13	Rule 100004 (Registry RunKey)
Defense Evasion	T1070.004	File Deletion	Sysmon Event ID 23	Rule 100005 (File Deletion)

7. Threat Hunting, Ingestion Metrics & Analytical Results

Over the testing period, the Wazuh SIEM platform ingested and correlated high volumes of telemetry across all endpoints:

Telemetry Stream	Source Node	Total Events Ingested	Key Ingested Types	Alert Severity Levels
Sysmon for Linux (eBPF)	Kali Linux (004)	10,135+ Events	Process Terminate (5), Process Create (1)	Level 3 to Level 10
Windows Sysmon & Events	Windows 11 (003)	1,427+ Events	Process Create (1), Network (3), Event 4625	Level 3 to Level 12
Linux PAM & Auth Logs	Ubuntu / Kali	1,044+ Events	Rule 5502 (PAM sudo session), sshd logs	Level 3 to Level 8
Total Lab Telemetry	All Multi-Nodes	12,600+ Events	Cross-platform unified log correlation	Unified SIEM Ingestion

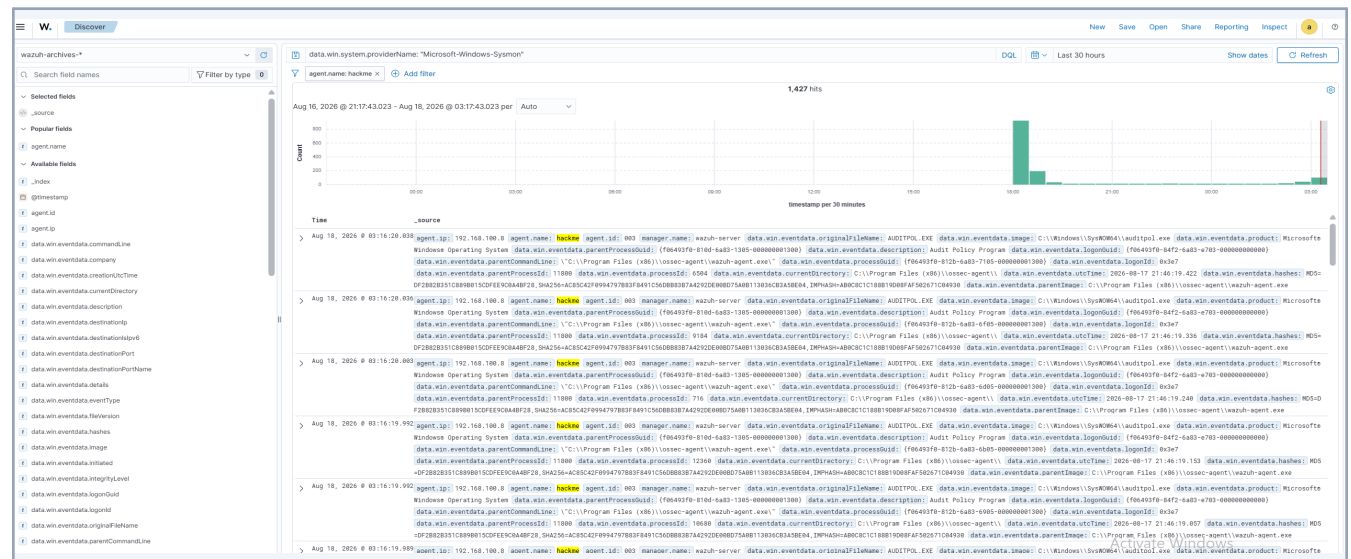


Figure: Wazuh Discover Ingestion of Over 1,400+ Windows Sysmon Process Events

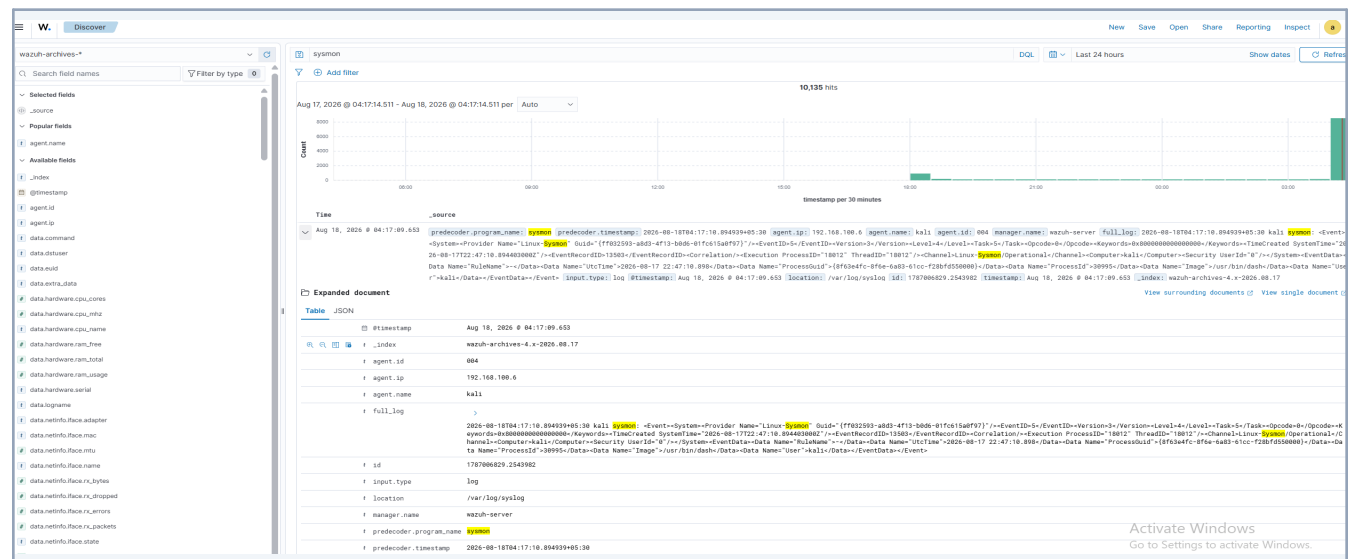


Figure: Wazuh Discover Ingestion of 10,135+ Linux Sysmon Events via eBPF Kernel Tracepoints

8. Security Alert Details & Forensic Investigation

When attack events trigger threshold conditions, Wazuh generates detailed alert records containing full metadata for forensic analysis:

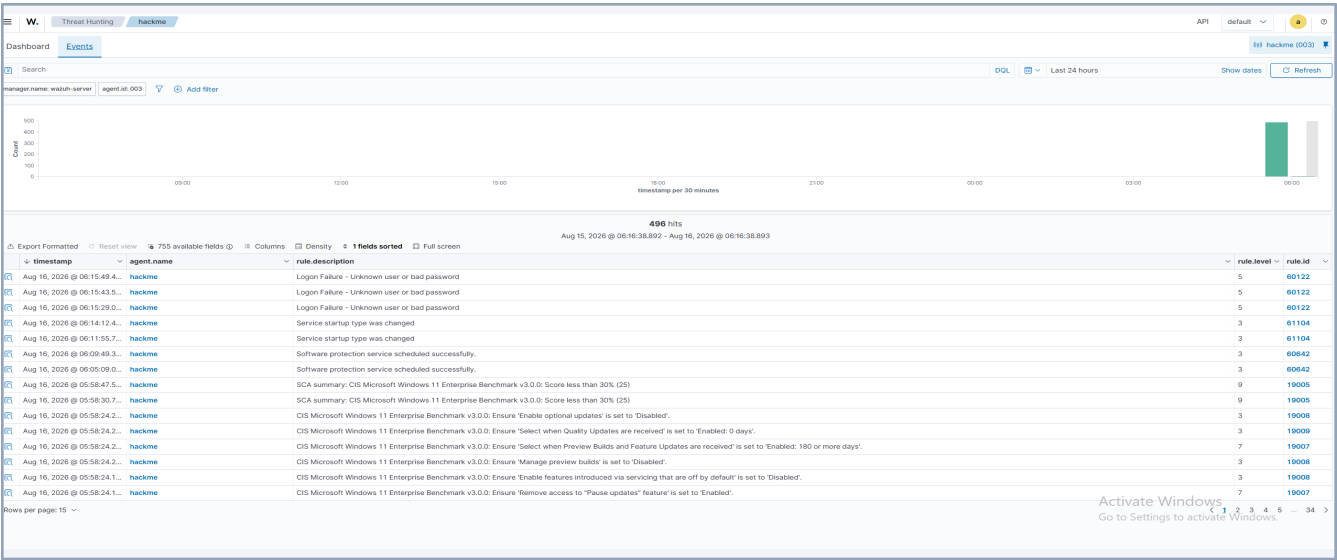


Figure: Wazuh Alert Detail View: Windows Logon Failure (Rule 18152 / Event ID 4625)

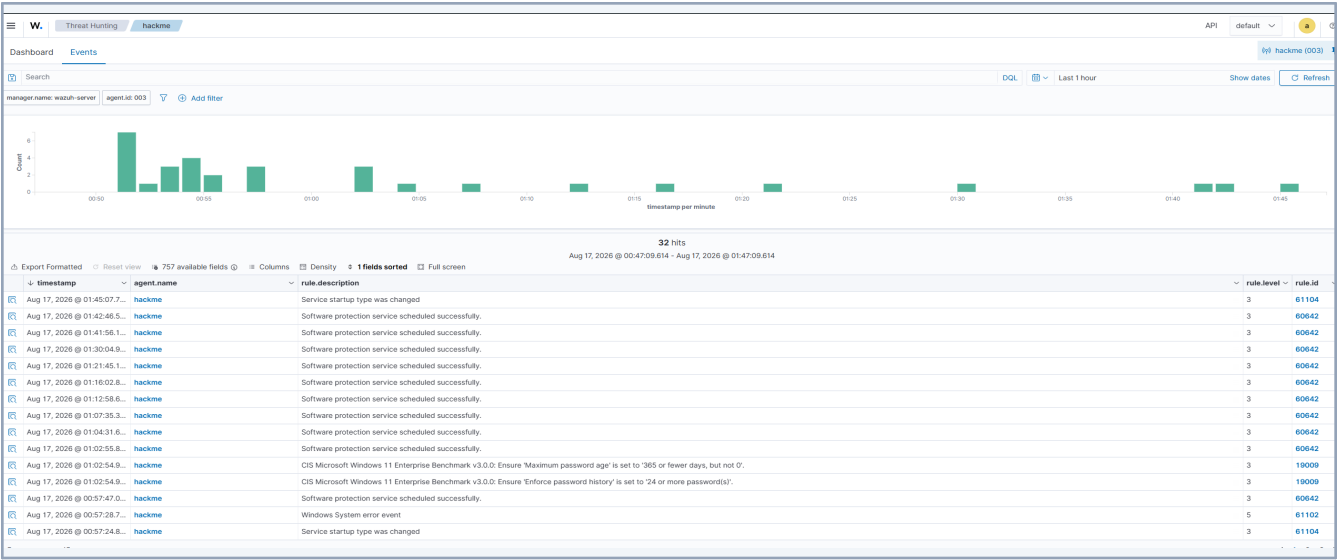


Figure: Real-Time Threat Hunting Events Table Showing Rule IDs, Levels, and CIS Benchmarks

9. Operational Troubleshooting & Hardening

Throughout the engineering lifecycle of this SOC lab, several operational challenges were diagnosed and resolved:

- **Firewall & Agent Connectivity:** Resolved agent connection timeouts by explicitly permitting TCP ports 1514 (log stream) and 1515 (enrollment) in Ubuntu UFW. Tested connectivity using `Test-NetConnection -Port 1514` from Windows.
- **Sysmon Event Channel Subscriptions:** Resolved missing Sysmon telemetry on Windows by adding the `Microsoft-Windows-Sysmon/Operational` channel with `eventchannel` inside `ossec.conf`.
- **Linux eBPF Kernel Compatibility:** Resolved Sysmon for Linux compilation warnings by confirming host Linux kernel version was `>= 5.4` with BPF filesystem mounted at `/sys/fs/bpf`.
- **Agent Service Daemon Recovery:** Created rapid recovery workflows (`NET START WazuhSvc` on Windows and `systemctl restart wazuh-agent` on Linux) to recover from system sleep/hibernation disconnections.

10. Conclusion & Future Roadmap

This project successfully demonstrated the end-to-end design, implementation, offensive simulation, and defensive monitoring of an enterprise-caliber Security Operations Center (SOC) home laboratory. By uniting Wazuh SIEM with Microsoft Sysmon and eBPF technology, the lab delivers actionable visibility across multi-stage cyber threats.

Future Enhancements Roadmap

- **Network Intrusion Detection (NIDS):** Integrate Suricata for deep packet inspection and network-level exploit detection.
- **Threat Intelligence & Enrichment:** Connect VirusTotal, AbuseIPDB, and AlienVault OTX APIs into Wazuh for automated IoC scoring.
- **SOAR Automation:** Deploy Shuffle SOAR to automate active response actions (such as automated endpoint network isolation upon critical alert triggers).
- **Incident Management:** Integrate TheHive and MISP for structured security incident ticketing and collaborative threat intel sharing.

Project Status: COMPLETED & VERIFIED	Lead Author: Natto Muni Chakma
Degree Program: B.Tech Computer Science & Eng.	Institution: Andhra University College of Eng.
Project Demonstration: https://youtu.be/duEibRGYMH0	GitHub Repository: github.com/NATTOMR